Optimal Siting and Sizing of Solar, Wind, and Battery Storage to Compensate for Nuclear Decommissioning in France

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Nuclear power is the backbone of France's electricity system. In 2024, 67.1% of French electricity came from nuclear plants, with 57 reactors operating across 18 sites and a total nuclear capacity of 62.9 GW, the highest nuclear share of any country in the world. This unique energy profile has allowed France to maintain one of the lowest carbon intensities in Europe.

However, this fleet is aging. The majority of France's reactors were connected to the grid between 1977 and 1993, and were originally designed for a 40-year operational lifetime. The average age of the current fleet is 39 years, and the oldest still-operating reactors date back to 1979. If a strict 40-year lifetime were applied with no new nuclear construction, the consequences would be substantial: based on IAEA PRIS data, approximately 7.75 GW of nuclear capacity would need to be retired by 2030, an additional 4 GW between 2031 and 2035, and a further 6 GW between 2036 and 2040, totaling roughly 18 GW of lost generation capacity by 2040. Meanwhile, France's share of wind and solar stands at only 13%, well below the EU average of 30%, leaving a large gap to fill.

This raises the following optimization problem: how much solar, wind, and battery storage would be required to compensate for this capacity loss, and where in France should these resources be located?

France is a very geographically diverse country with solar irradiance substantially higher in the south than in the north, wind resources concentrated along coastal and elevated regions, and large portions of the territory are constrained by protected ecosystems, forests, agricultural land, and urban areas. Beyond resource potential, the temporal mismatch between solar and wind production and peak electricity demand requires explicitly co-optimizing storage capacity alongside generation. Additionally, connecting new capacity to the grid is cheaper in some locations than others, which further influences where it makes sense to invest. This project will address all three dimensions simultaneously through a grid-based spatial optimization model covering all of France.